CYBERSECURITY FUNDAMENTALS

A Simple Guide to Ethical Hacking & Online Safety

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CYBER SECURITY INTERN • SHAHBAZ SECURITY

DAY 01 – DAY 02 | LEARNING REPORT

01 | WHAT IS CYBERSECURITY?

Cybersecurity means keeping computers, networks, websites, and personal information safe from online attacks. It helps stop people from stealing data, changing information, or disrupting digital services.

02 | THE CIA TRIAD

KEY IDEA	IN SIMPLE WORDS	REAL-LIFE EXAMPLE
Confidentiality	Only the right people can see private information.	A banking password helps keep an account private.
Integrity	Information stays correct and is changed only with permission.	A bank protects transaction records from unauthorized changes.
Availability	Systems and information are ready when people need them.	Backups help a website recover if its main server fails.

03 | THREE TYPES OF HACKERS

WHITE HAT	Works with permission to find security problems and help fix them.
BLACK HAT	Breaks into systems without permission to steal, damage, or misuse information.
GREY HAT	May find security problems without permission, but does not always have harmful goals.

04 | FIVE PHASES OF ETHICAL HACKING

1. Reconnaissance	Collect information about the target before testing.
2. Scanning	Look for active devices, open ports, services, and possible weaknesses.
3. Gaining Access	Check whether a weakness could allow access, only within the approved test scope.
4. Maintaining Access	Check whether access could continue and whether security tools can detect it.
5. Covering Tracks	Learn how attackers hide activity. Ethical testers keep clear records, protect logs, and clean up test files with permission.

05 | PENETRATION TESTING VS. VULNERABILITY ASSESSMENT

Vulnerability assessment looks for security weaknesses and ranks them by risk. Penetration testing safely tests selected weaknesses to see what an attacker might be able to do. One mainly finds and prioritizes issues; the other checks their real-world impact.

06 | CYBERSECURITY CAREER PATHS

People in cybersecurity can work as penetration testers, security analysts, security engineers, incident responders, digital forensics analysts, or security consultants. These jobs help find risks, protect systems, investigate problems, and improve security.

07 | WHAT I LEARNED

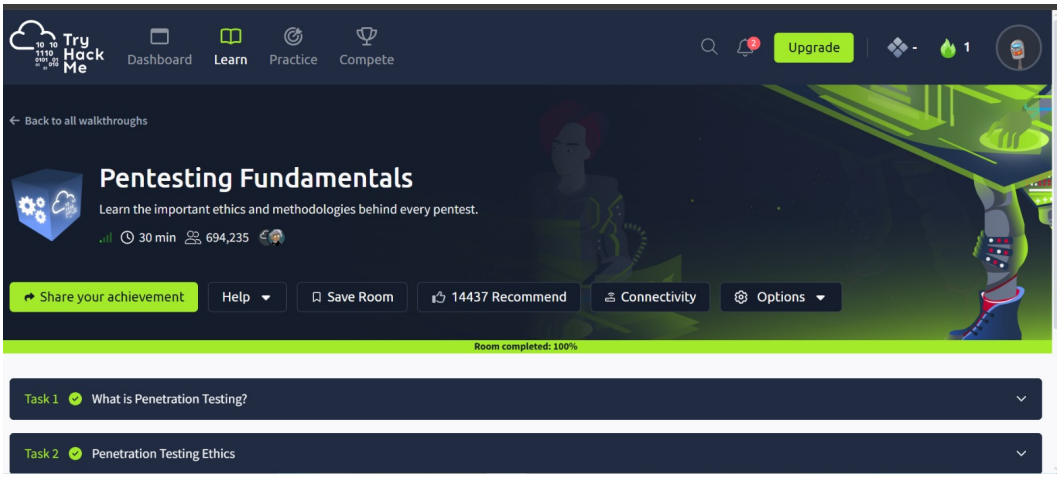
For this activity, I completed three TryHackMe learning rooms: Pentesting Fundamentals, Offensive Security Intro, and Careers in Cyber. The screenshots on the next page show 100% completion for each room. I learned the basic ideas behind cybersecurity, how ethical hackers work, and why testing must always have permission and clear rules.

MY KEY TAKEAWAY
Ethical hacking means finding security problems in a responsible way and helping make digital systems safer.

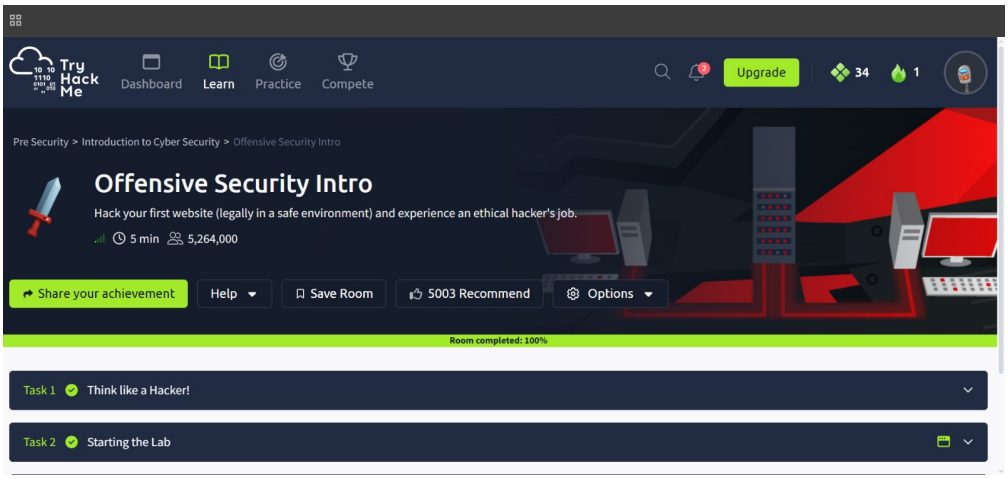
PRACTICAL LEARNING EVIDENCE

TRYHACKME • COMPLETION SCREENSHOTS

01 PENTESTING FUNDAMENTALS



02 OFFENSIVE SECURITY INTRO



03 CAREERS IN CYBER

